SMPC PLANT INTERLOCK (TA UNIT)

TA No.1

TP-102A/B STOP (SD-1102)

If TP-102A/B stops for any cause, there will be a back flow of the liquid in the reactor, causing excess pressure in TD-102A or clogging of the RQ feed line. To prevent this, the RQ feed valve (FICV-1102) is closed if both TP-102A and TP-102B stop.

CATA ANALYZER FEED STOP (SD-1100)

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Since the film at the X-ray irradiation section of the analyzer flow cell is weak in strength, any rise in the pressure in the line due to clogging of the sample line or any other cause is detected by PIS-1111, which in turn cuts off the feed to the analyzer flow cell, thus preventing damage to the film. This is also the case when the gas detector (AI-1007) in the analysis room is activated.

TD-203 HIGH PRESS (SD-1203)

If the pressure (PIS-1205) of TD-203 rises above 0.15K, TZ-201 is closed, so that the removal of TA slurry from the reactor, which is the source of pressure. The purpose of this is to prevent a rise in the internal pressure of TD-203 (D.P. 1.8 kg/cm^2G) due to clogging of the TE-203A/B tube or TM-203 or any other cause.

TD-203 RUPTURE DISK ON (SD-1204)

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A rupture disk is installed to prevent the destruction of TD-203. If the temperature on the outlet side (TIS-1219) rises above the set value at the time of the activation of the rupture disk, CCW is supplied to TD-206 to prevent acetic acid vapor from being discharged to the atmosphere.

REACTOR SD (SD-1201)

(1) The O_2 concentration in waste gas is continuously measured by three units of O_2 analyzer (AIS-1201, 1202, 1203). If two of the three analyzers show a value higher than the set value (high O_2), if the air feed rate falls below the set value (FICS-1201 low flow), or if TP-201A/B stops and the TD-201 reflux ratio (FI-1220 low flow) lowers, the reaction air valves (FICV-1209A to D, SKV-1201A to D), SDV-1202 and FICV-1210 are closed, so that the pressure and the liquid are held.

(2) If the reactor is shut down, the compressor, steam turbine and generator turbine will be brought to a shutdown and an HPVGT shutdown.

(3) The following operations are automatically carried out as the steps to be taken after the shutdown of the reactor:

a. The PX feed valve (FV-1101), TP-101 discharge valve (KCV-1101) and TP-201A/D discharge valves (KCV-1201, 1202) are closed, and TP-101A, TP-201A/D, TP-104A/B and TP-105A are stopped.

b. BW feed to TE-201B/C is stopped to prevent the no-load operation of the pump (TP-1131) due to the low level of TE-1134. (LICV-1222, 1223 are closed.)

c. The destination of the discharge from the TT-202 upper side and TT-203B is switched from TD-204 over to the sewer.

d. The steam valves (SDV-3104, 3105) of TE-1132 and TE-1133 are closed.

e. The TD-201 side cut (FICV-1215), TD-202 sewer (FICV-1203) and TE-202 sewer (LICV-1210) are closed.

f. TE-202 outlet water recovery is switched over to DIW. (KCV-1515 is opened, and KCV-1514 is closed.)

TA No.2

REACTOR LOW PRESS (SD-1202)

If the reactor inside pressure falls sharply, a sharp increase in the O_2 concentration in the reactor will take place due to a rapid evaporation of solvent acetic acid or a flow of excess air supply into the reactor, bringing the O_2 concentration into the explosive range. To prevent this, SDV-1202 is closed if the internal pressure in the reactor (PICS-1202) falls below the set value, so that a pressure drop will not occur.

TE-201A, B, C HIGH LEVEL (SD-1220)

If the actual liquid level on the shell side of TE-201A, B, C rises due to a breakdown of the level gauge (LIC-1221, 1222, 1223) of TE-201A, B, C, mist is entrained in the generated steam, adversely affecting the stream turbine. Because of this, if LAS of TE-201 is activated, LICV is closed, so that the feed of 1BW to TE-201 is stopped.

TM-300A, B, C SD (SD-1300A/B/C)

If the seal coolant pressure, the spindle oil pressure or the gear box pressure falls below the set value and if the torque limiter is activated, the centrifuge is stopped, and the feed LICV is closed.
If the centrifuge stops, the separation of TA and ML becomes impossible, and ML drops into TD-300, causing a deterioration in the quality of TA. To prevent this, the feed LICV is closed if the main motor stops.

TM-301A/B HIGH LEVEL (SD-1301A/B)

If the level in the pan rises due to overfeeding to TM-301A/B or clogging of the overflow line to TD-300 in the pan, the slurry flows onto the dryer side, resulting in deteriorated product quality or increased load on the dryer.
Therefore, in the case of LIS-1304 HIGH LEVEL of the TM-301 pan, the feed to TM-301 is stopped and the cake washing FDQ is closed.

TM-301A/B HIGH & LOW PRESS (SD-1303A/B)

When TM-301 is maintained or repaired, it is separated from the dryer system.
In the case of TM-301 HIGH or LOW PRESS, SDV is opened to prevent damage to the casing caused by excess pressure by gland N_2 or a negative pressure due to leak from the filtrate KCV.

TD-302 HIGH LEVEL (SD-1302)

In the case of TD-302 HIGH LEVEL due to trouble in TP-302 or the clogged SUC

& discharge line, the TM-301 filtrate suction line (between TM-301 and TD-302) is clogged up with the filtrate liquid (because of high level of TD-302), leading to a lowered flow rate of the blow back gas or clogging of the tube with TE-302 TA. Therefore, if the level of TD-302 exceeds the setting, the TM-301 vacuum is cut.

Dryer SD (SD-1304)

In the case of trouble in the dryer system or the related equipment, if any equipment is stopped due to a certain cause to prevent deterioration of the product quality caused by damage to the equipment, clogging or a drying failure, all the equipment on the upstream side of the relevant equipment is stopped.
If the flow rate of the drying N_2 lowers or TC-301 stops, the dryer is shut down and the pneumatic transfer is stopped.
For RVF, the RVF ESD SW is set to ON and the slurry treatment is stopped.

TA No.3

TT-301 HIGH LEVEL (SD-1305)

If the level of TT-301 rises due to the clogged TP-310 SUC & DEL line or a pump breakdown and TT-301 HIGH LEVEL occurs to prevent inflow of acetic acid to the dryer side, the TT-301 top spray FDQ is stopped.
The TM-302 → TT-301 FDQ purge sequence is also stopped.
When TT-301 returns to the normal level (50%), the interlock is automatically released and the top spray is placed in auto control.

TC-302 STOP (SD-1306)

When TC-302 is run and the dryer outlet TA is pneumatically transferred to the product silo (TTK-400, 401), the product silo may be placed under a negative pressure. The negative pressure in the silo causes damage to the silo and excess pressure of the TC-302 suction causes damage to TC-302.
Therefore, if TC-302 SUCTION PRESS HIGH or LOW PRESS occurs, TC-302 is stopped.

TC-302 STARTING CONDITION (SD-1306)

TC-302 can be started when KCV-1402A (to atmosphere) at the TM-403 outlet is closed and KCV-1402B at the TC-302 suction is opened, or when KCV-2801 is set to the PTK-800 side and KCV-2883 is set to the PTK-810 and 820 sides.

TM-501, 502 SD (SD-1501)

If TM-501 or TM-502 is stopped for protection of the equipment, the TM-501 ML feed valve is closed (FICV-1311), and the 10S feed valve (FICV-1551) and the TM-502 25S feed valve (FICV-1552) are closed up to near the fully closed position (MV = 5%).

TM-503 SD (SD-1503)

If the spindle oil pressure and the gear box pressure fall below the settings to prevent damage to the centrifuge and if the torque limiter is activated, the centrifuge is stopped and the feed valve (LICV-1505) and the feed tube purge valve (KCV-1584) are closed.

TA SILO SD (SD-1400A/B)

If the flow rate of the pneumatically transferred gas falls below the setting to prevent clogging of the pneumatic transfer line, the damper valves (HZSV-1400, 1401) are closed and the rotary valves (TZ-400, 401, 411 and 412) are stopped.

PROTECT MISOPERATING FOR TA SILO (SD-1401)

If the air is sucked from the TC-302 suction due to an operation error of KCV, there is a possibility of dust explosion in the silo.
Therefore, the preventive mechanism against operation errors is installed so that the operation of KCV-1402A/B is disabled during operation of TC-302.

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TA No.4

TD-1201 HIGH LEVEL (SD-3201)

In the flush drums, the surplus of 10SC and 3SC is transferred sequentially from TD-1210 → TD-1203 → TD-1201 depending on the differential pressure. If the liquid transfer is disabled due to trouble in each SC pump, 10SC and 3SC are transferred to TD-1201, which is then filled with SC to the brim, and SC enters the 1S header. To prevent this, SDV-3210 is opened if the TD-1201 level exceeds the setting and SC is discharged to the process draining trench.
When the TD-1201 level returns to the normal level (50%), the interlock is automatically released and SDV-3210 is closed.

HPVGT SD (SD-1901)

If the flow rate of WG to TF-901 becomes low, the tube may be damaged due to abnormal heating of the TF-901 heating tube. Therefore, if FICS-1901 LOW FLOW occurs, all the main and pilot burners are extinguished. In the case of TIS-1905 and TDIS-1906 HIGH TEMP, FG LOW PRESS (PS-1909) and REACTOR SD, all the burners are also extinguished.
If the above interlock is activated, the TT-202 outlet WG is switched from the HPVGT side to the bypass side.